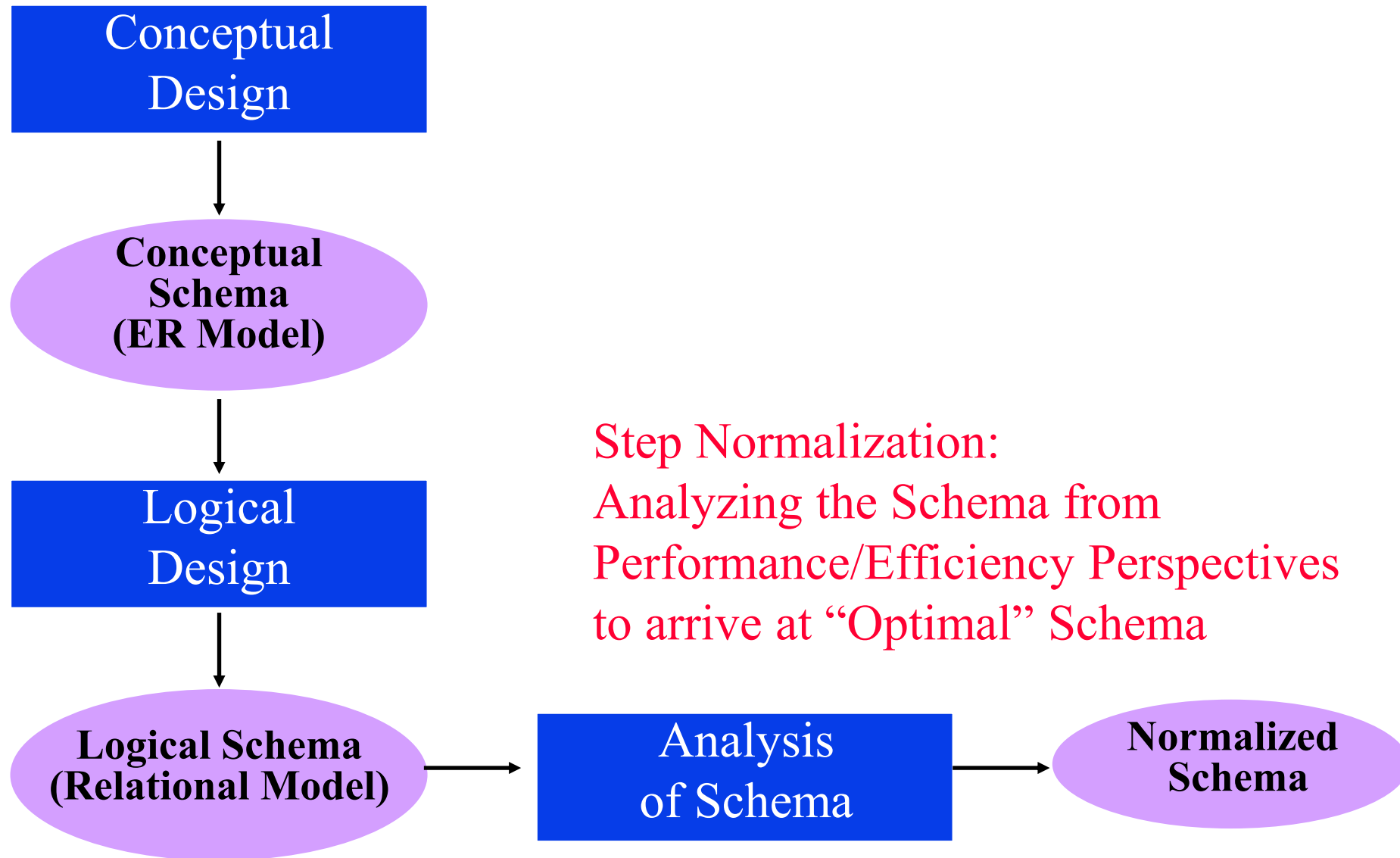


Functional Dependencies & Normalization

Design Process



Functional dependencies

- Functional dependencies (FDs) are used to specify *formal measures* of the "goodness" of relational designs
- FDs and keys are used to define **normal forms** for relations
- FDs are **constraints** that are derived from the *meaning* and *interrelationships* of the data attributes
- A set of attributes *X functionally determines* a set of attributes Y if the value of X determines a unique value for Y
- $X \rightarrow Y$ holds if whenever two tuples have the same value for X, they *must have* the same value for Y
- For any two tuples t1 and t2 in any relation instance r(R): *If* $t1[X]=t2[X]$, *then* $t1[Y]=t2[Y]$

Functional dependencies

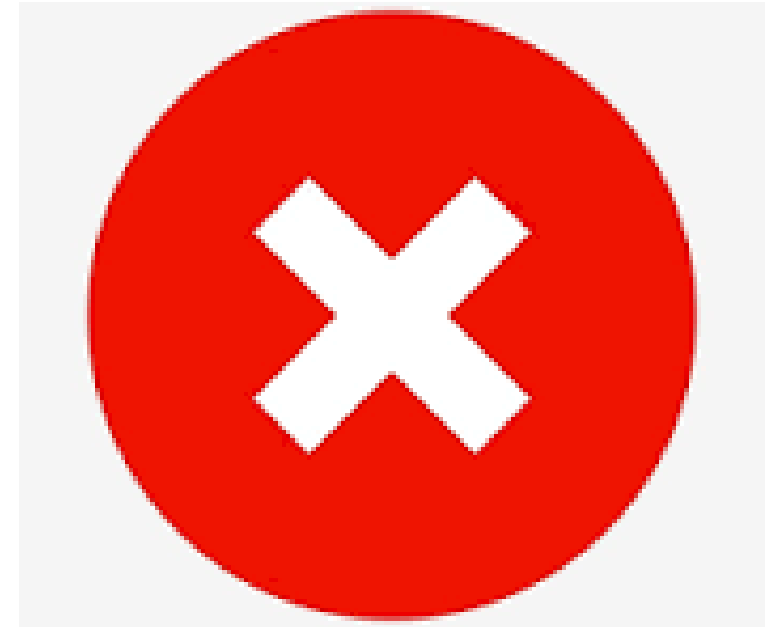
ENo	ENAME	ETitle	PNo	PName	PBudget	EPDuration	EPResponsibility
1	Kamandanu	SoftwareEngineer	L983	Finance Project	89000	546	Technical Lead

is this the best form of relational model?

Functional dependencies

ENo	ENAME	ETitle	PNo	PName	PBudget	EPDuration	EPResponsibility
1	Kamandanu	SoftwareEngineer	L983	Finance Project	89000	546	Technical Lead

- ENo $\rightarrow$ Ename
- ENo $\rightarrow$ Etitle
- PNo $\rightarrow$ Pname
- PNo $\rightarrow$ Pbudget
- {ENo,PNo} $\rightarrow$ EPduration
- {ENo,PNo} $\rightarrow$ EPResp



Relational Model

EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

WORKS

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000
P5	CAD/CAM	500000



Guideline 1: Semantics of the Relation Attributes

GUIDELINE 1: Informally, **each tuple in a relation should represent one entity or relationship instance.** (Applies to individual relations and their attributes).

- Attributes of different entities (EMPLOYEEs, DEPARTMENTs, PROJECTs) should not be mixed in the same relation
- Only foreign keys should be used to refer to other entities
- **Entity and relationship attributes should be kept apart as much as possible.**

1. Semantics of the Relation Attributes

Sample

ENo	ENAME	Etitle	PNo	PName	PBudget	WDuration	WResp
E01	Trump	Accountant	P01	Project1	90,000.00	100	Financial

- **Semantic** of attributes are **mixed**
Name of employee in the same tuple with the name of project
- **what can be concluded?**
- **does it meet the criteria?**

Guideline 2: Redundant Information in Tuples and Update Anomalies

GUIDELINE 2:

Design a schema that does not suffer from the insertion, deletion and update anomalies.

If there are any present, then note them so that application can be made to take them into account.

2. Redundant Information in Tuples and Update Anomalies

- Mixing attributes of multiple entities may cause problems
- **Information is stored redundantly wasting storage**
- Problems with update anomalies:
 - Insertion anomalies
 - Deletion anomalies
 - Modification anomalies

2. Redundant Information in Tuples and Update Anomalies

ENo	ENAME	ETitle	PNo	PName	PBudget	WDuration	WResp
E01	Trump	Accountant	P01	Project1	90,000.00	100	Financial

Sample Redundant

- If 300 employees work in Project1, then all attributes of project 1 such as pno, budget will be written multiple times
- If need to change the project name, must change all tuples
- Inserting employee, must also insert project
- Deleting project will also delete part of employee.

Guidelines 3: Null Values in Tuples

GUIDELINE 3: Relations should be designed such that their tuples **will have as few NULL values as possible**

- Attributes that are NULL frequently could be placed in separate relations (with the primary key)
- **Reasons for nulls:**
 - attribute not applicable or invalid
 - attribute value unknown (may exist)
 - value known to exist, but unavailable

3. Null Values in Tuples

ENo	ENAME	ETitle	PNo	PName	PBudget	WDuration	WResp
E01	Trump	Accountant	P01	Project1	90,000.00	100	Financial

Sample

- In case pno is not PK, only input info of employee then more info of project would be null

Guideline 4: Spurious Tuples

GUIDELINE 4:

- Bad designs for a relational database may result in erroneous results for certain JOIN operations
- The "lossless join" property is used to guarantee meaningful results for join operations
- There are **two important properties of decompositions**:
 - non-additive or losslessness of the corresponding join
 - preservation of the functional dependencies.

4. Spurious Tuples

Note that property (a) is extremely important and *cannot* be sacrificed.
Property (b) is less stringent and may be sacrificed.

ABC Ename 	123 Ewduration 
Biden	90
Trump	100

ABC Eno 	ABC Ename 	ABC Etitle 	ABC Epo 	ABC Epname 	123 Epbudget 
E01	Trump	Accountant	P01	Project1	100
E02	Biden	SW Engineer	P01	Project1	10,000

4. Spurious Tuples

Where is spurious tuples?

ABC Ename 	123 Ewduration 
Biden	90
Trump	100

ABC Eno 	ABC Ename 	ABC Etitle 	ABC Epo 	ABC Epname 	123 Epbudget 
E01	Trump	Accountant	P01	Project1	100
E02	Biden	SW Engineer	P01	Project1	10,000

Normalization of Relation

- **Normalization:** The process of **decomposing unsatisfactory "bad" relations by breaking up** their attributes into smaller relations.
- **Normal form:** Condition **using keys and FDs of a relation to certify** whether a relation schema is in a particular normal form

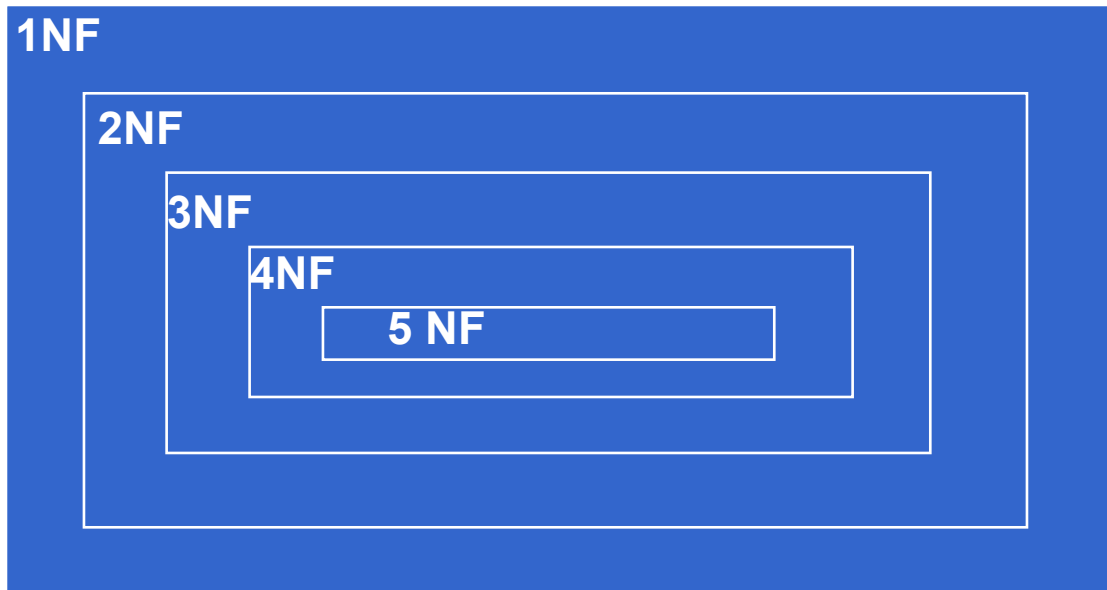
Normalization of Relation

Purpose :

1. Reduce data duplication
2. Ensure data is in the right table
3. Improve data management efficiency save storage space
4. Improve performance
5. Improve security
6. Ease maintenance
7. Ensure the database is working optimally

Normal Form

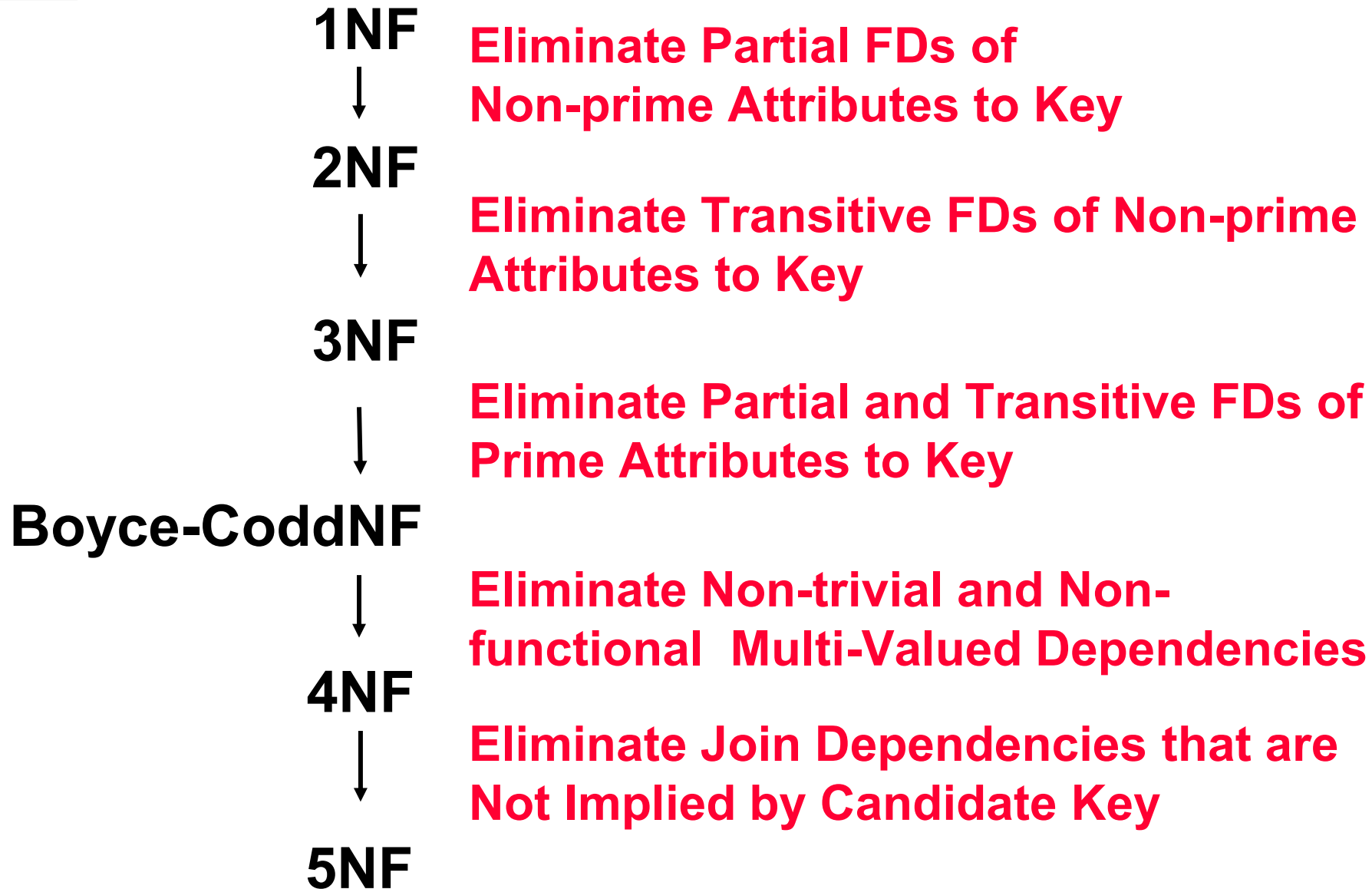
- A Normal Form is a Condition using Keys and FDs to Certify Whether a Relation Schema meets Criteria
 - Primary keys (1NF, 2NF, 3NF)
 - All Candidate Keys (2NF, 3NF, BCNF)
 - Multivalued Dependencies (4NF)
 - Join Dependencies (5NF)



Normal Form

- **1NF** based on definition of relation
- **2NF, 3NF, BCNF** based on keys and FDs of a relation schema
- **4NF** based on keys, multi-valued dependencies : MVDs;
- **5NF** based on keys, join dependencies : JDs
- **Additional properties** may be needed to ensure a good relational design (lossless join, dependency preservation)

Normal Form



Practical use Normal Form

- **Normalization** is carried out in practice so that the **resulting designs are of high quality and meet the desirable properties.**
- The practical utility of these normal forms **becomes questionable** when the constraints on which they are **based are hard to understand or to detect.**
- The database designers ***need not* normalize to the highest possible normal form.** (usually up to 3NF, BCNF or 4NF).
- **Denormalization:** the process of **storing the join of higher normal form relations as a base relation**—which is in a lower normal form

Practical use Normal Form

- If a relation schema has more than one key, each is called a **candidate key**. One of the candidate keys is *arbitrarily* designated to be the **primary key**, and the others are called *secondary keys*.
- A **Prime attribute** must be a member of *some candidate key*
- A **Nonprime attribute** is not a prime attribute—that is, it is not a member of any candidate key.

1NF

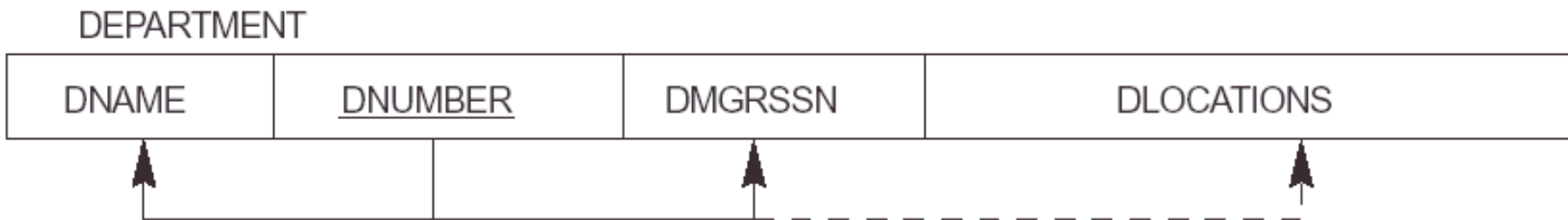
- Disallows composite attributes, multivalued attributes, and **nested relations**;
- attributes whose *values for an individual tuple* are non-atomic
- Considered to be part of the definition of relation

1NF

- All Attributes **Must Be Atomic Values**:
 - Only **Simple and Indivisible** Values in the Domain of Attributes.
 - Each Attribute in a 1NF Relation is a **Single** Value
 - Disallows Composite Attributes, Multivalued Attributes, and Nested Relations (Non-Atomic)
- 1NF Relation **cannot have an Attribute Value** :
 - A Set of Values (**Set-Value**)
 - A Tuple of Values (**Nested Relation**)
- 1NF is a Standard Assumption of Relation DBs

Normalitation into 1NF (case-1)

- Consider Following Department Relation
- What is the Inherent Problem?



DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

Normalitation into 1NF (case-2)

SSN	ENAME	PROJS	
		PNUMBER	HOURS

SSN	ENAME	PNUMBER	HOURS
123456789	Smith,John B.	1	32.5
		2	7.5
666884444	Narayan,Ramesh K.	3	40.0
453453453	English,Joyce A.	1	20.0
		2	20.0
333445555	Wong,Franklin T.	2	10.0
		3	10.0
		10	10.0
		20	10.0
999887777	Zelaya,Alicia J.	30	30.0
		10	10.0
987987987	Jabbar,Ahmad V.	10	35.0
		30	5.0
987654321	Wallace,Jennifer S.	30	20.0
		20	15.0
888665555	Borg,James E.	20	null

Possible Solutions (case-1)

- **Decompose:** Move the Attribute DLOCATIONS that Violates 1NF into a Separate Relation
DEPT_LOCATIONS(DNUMBER, DLOCATION)
- Expand the key to have a Separate Tuple in the DEPARTMENT relation for each location (below)

DNAME	<u>DNUMBER</u>	DMGRSSN	<u>DLOCATION</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

Possible Solutions (case-2)

SSN	ENAME	PROJS	
		PNUMBER	HOURS

SSN	ENAME	PNUMBER	HOURS
-----	-------	---------	-------

123456789	Smith,John B.	1	32.5
		2	7.5
666884444	Narayan,Ramesh K.	3	40.0
453453453	English,Joyce A.	1	20.0
		2	20.0
333445555	Wong,Franklin T.	2	10.0
		3	10.0
		10	10.0
		20	10.0
999887777	Zelaya,Alicia J.	30	30.0
		10	10.0
987987987	Jabbar,Ahmad V.	10	35.0
		30	5.0
987654321	Wallace,Jennifer S.	30	20.0
		20	15.0
888665555	Borg,James E.	20	null

EMP_PROJ1

<u>SSN</u>	ENAME
------------	-------

EMP_PROJ2

<u>SSN</u>	<u>PNUMBER</u>	HOURS
------------	----------------	-------

Question

ABC Eno 	ABC Ename 	ABC Etitle 	ABC Epo 	ABC Epname 	123 Epbudget 	123 Ewduration 	ABC Ewresp 
E01	Trump	Accountant	P01	Project1	90,000	100	Financial

Is 1NF??

Uses the concepts of **FDs**, **primary key**

Definitions:

- **Prime attribute** - attribute that is member of the primary key K
- **Full functional dependency** - a FD $Y \rightarrow Z$ where removal of any attribute from Y means the FD does not hold any more

2NF

EMP_PROJ1

<u>SSN</u>	ENAME
------------	-------

EMP_PROJ2

<u>SSN</u>	<u>PNUMBER</u>	HOURS
------------	----------------	-------

Examples:

- {SSN, PNUMBER} → HOURS is a full FD since neither SSN → HOURS nor PNUMBER → HOURS hold
- {SSN, PNUMBER} → ENAME is *not* a full FD (it is called a *partial dependency*) since SSN → ENAME also holds

2NF

- A relation schema R is in **second normal form (2NF)** if every non-prime attribute A in R is fully functionally dependent on the primary key
- R can be decomposed into 2NF relations via the process of 2NF normalization

2NF

- Second Normal Form Focuses on the Concepts of Primary Keys and Full Functional Dependencies
- Intuitively:
 - A Relation Schema R is in **Second Normal Form (2NF)** if Every Non-Prime Attribute A in R is Fully Functionally Dependent on the Primary Key
 - R can be Decomposed into 2NF Relations via the Process of 2NF Normalization
 - Successful Process Typically Involves **Decomposing R into Two or More Relations**
 - Iteratively **Applying to Each Relation in Schema**

Nested Relations

Tabel Karyawan:

KaryawanID	Nama	Jabatan	DepartemenID
1	Ali	Developer	1
2	Budi	Manager	1
3	Citra	Designer	2

Tabel Departemen:

DepartemenID	NamaDepartemen	ManajerID
1	IT	2
2	Marketing	3

Tabel Proyek:

ProyekID	NamaProyek	DepartemenID	KaryawanID
1	Proyek A	1	1
2	Proyek B	2	3

Normalization into 2NF

ABC Eno 	ABC Ename 	ABC Etitle 	ABC Epo 	ABC Epname 	123 Epbudget 	123 Ewduration 	ABC Ewresp 
E01	Trump	Accountant	P01	Project1	90,000	100	Financial

- Determine Entitas!
- Determine Fully FD!

Normalization into 2NF

ABC Eno 	ABC Ename 	ABC Etitle 	ABC Epo 	ABC Epname 	123 Epbudget 	123 Ewduration 	ABC Ewresp 
E01	Trump	Accountant	P01	Project1	90,000	100	Financial

- Eno → Ename, Etitle (**Employee**)
- Epo → Epname, Epbudget (**Project**)
- {Eno, Epo} → Ewduration, Ewresp (**Works**)

3NF

- Third Normal Form Focuses on the Concepts of **Primary Keys and Transitive Functional Dependencies**
- Intuitively:
 - A Relation Schema R is in **Third Normal Form (3NF)** if it is in 2NF *and* no Non-Prime Attribute A in R is Transitively Dependent on Primary Key
 - R can be Decomposed into 3NF Relations via the Process of 3NF Normalization
- In $X \rightarrow Y$ and $Y \rightarrow Z$, with **X as the Primary Key**, there is only a problem only **if Y is not a candidate key**.
EMP(SSN, Emp#, Salary), SSN $\rightarrow$ Emp# $\rightarrow$ Salary isn't Problem Since Emp# is a Candidate Key

3NF

ABC Eno 	ABC Ename 	ABC Etitle 
---	---	--

ABC Epo 	ABC Epname 	123 Epbudget 
---	--	--

ABC Eno 	ABC Epo 	123 Ewduration 	ABC Ewresp 
---	---	--	--

- EPo -> EPName and EPName->EPBudget
- Epo -> Epname -> Epbudget
- **Problem: Epname is not a prime key**

3NF

- EP1(Pno, Pname)
- EP2(Pname, Pbudget)

ABC Epo	ABC Epname
---------	------------

ABC Epname	123 Epbudget
------------	--------------

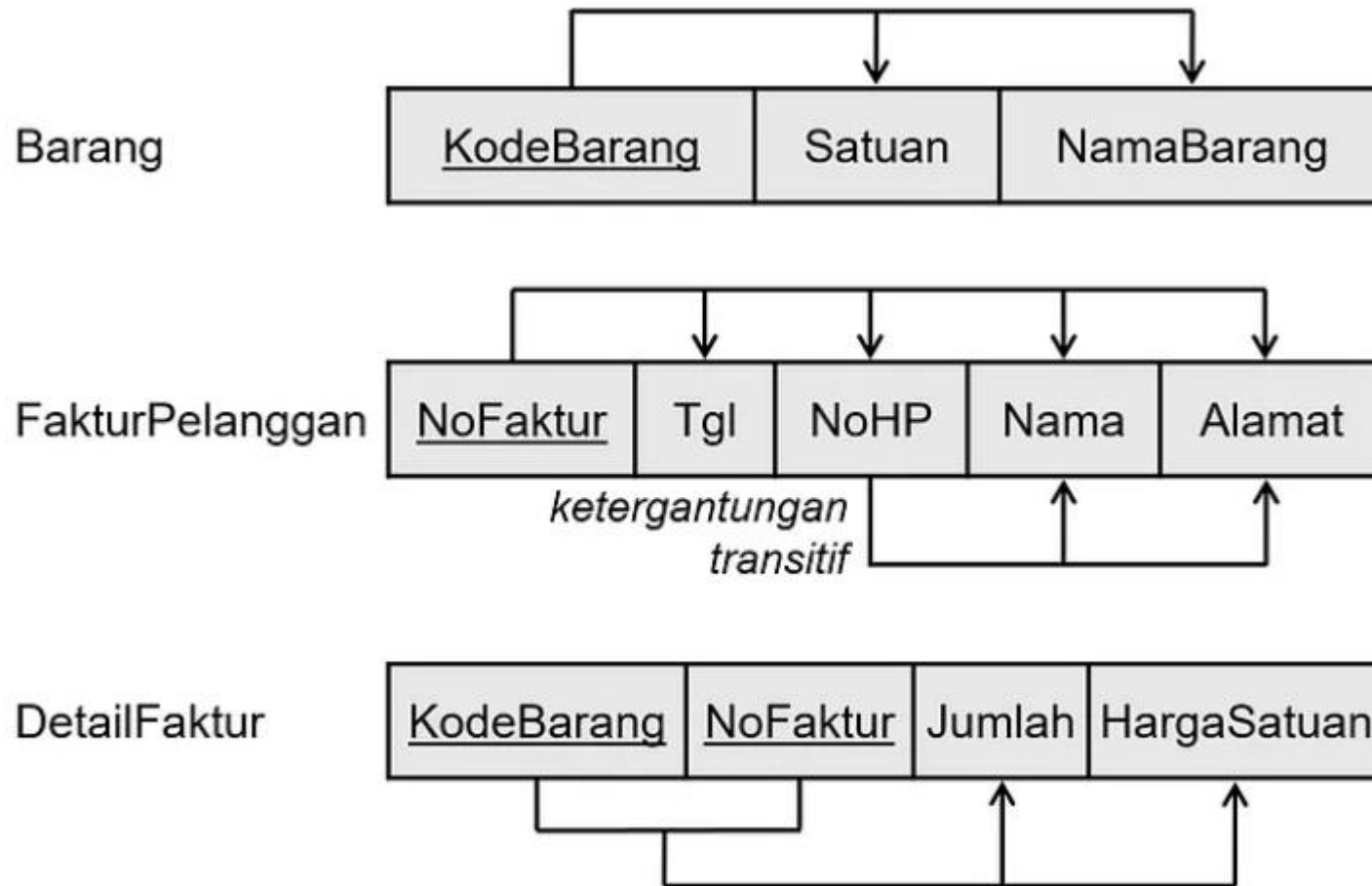
- **Alike 3rd N but..**
- A relation schema R is in Boyce-Codd Normal Form (BCNF) if whenever an FD $X \rightarrow A$ holds in R , then X is a superkey of R
- The goal is to have each relation in BCNF (or 3NF)
- Usually: in many cases Normalization is ended at BCNF

BCNF

ABC Ename ↑↓	ABC Etitle ↑↓	ABC Eresp ↑↓
--------------	---------------	--------------

- Ename and Etitle are Prime Key
- Eresp → Etitle
- Possible BCNF normalization:
 - E1 (Ename, Eresp) E2 (Etitle, Eresp)
 - E1 (Eresp, Etitle) E2 (Ename, Etitle)
 - E1 (Eresp, Etitle) E2 (Eresp, Ename)

BCNF



BCNF

NoTes	Tanggal	Waktu	NRP	Ruang
05.012	19-08-2005	08.00–09.30	99.005	205
05.034	19-08-2005	10.00–11.30	99.005	205
05.056	19-08-2005	10.00–11.30	99.013	304
05.034	23-08-2005	08.00–09.30	99.005	205
05.056	23-08-2005	10.00–11.30	99.005	205

- Solution?

Multivalue Dependency

- Focused on the Concept of Multi-Valued Dependencies
- A MVD $X \twoheadrightarrow Y$ Indicates that a Value of X Corresponds to Multiple Values of Y
- Consider EMP with MVDs:
 - $ENAME \twoheadrightarrow PNAME$ (E works on many Project)
 - $ENAME \twoheadrightarrow DNAME$ (E has many Dependents)

EMP

<u>ENAME</u>	PNAME	<u>DNAME</u>
Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John

4NF

- A Relation Schema R is in **Fourth Normal Form (4NF)** w.r.t Dependencies F (FD and MVD) if for every **Non-Trivial** MVD $X \twoheadrightarrow Y$ in F^+ , X is a Superkey for R\
- MVD $X \twoheadrightarrow Y$ in R is called trivial if
 - Y is subset of X, or
 - $X \cup Y = R$
- Reconsider EMP with MVDs:
 - $ENAME \twoheadrightarrow PNAME$ (E works on many P)
 - $ENAME \twoheadrightarrow DNAME$ (E has many Dependents)
- ENAME is Not a Superkey of R since Need Triple of ENAME, PNAME, and DNAME to Distinguish
- We need to Decompose EMP!

4NF

EMP

<u>ENAME</u>	PNAME	<u>DNAME</u>
Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John

EMP_PROJECTS

<u>ENAME</u>	<u>PNAME</u>
Smith	X
Smith	Y

EMP_DEPENDENTS

<u>ENAME</u>	<u>DNAME</u>
Smith	John
Smith	Anna

4NF

ABC Ename 	ABC Etitle 	ABC Eresp 
Susi	Analyst	DB Administrator
Susi	Analyst	Project Manager
Susi	Programmer	DB Administrator
Susi	Programmer	Project Manager

ABC Ename 	ABC Etitle 
Susi	Analyst
Susi	Programmer

ABC Ename 	ABC Eresp 
Susi	DB Administrator
Susi	Project Manager

5NF

- Alike 4th but ...
- Avoiding redundancy because join dependency. It is a like generalized multi
- Split into several table then join again, if the result is the same with before split then it satisfies 5th N
- Otherwise, must be normalized

5NF

ABC Ename	ABC Etitle	ABC Eresp
Budi	Accountan	DB Administrator
Busi	Analyst	DB Administrator
Busi	Manager	Project Manager
Susi	Analyst	DB Administrator
Susi	Programmer	DB Administrator

ABC Ename	ABC Etitle
Budi	Accountan
Busi	Analyst
Busi	Manager
Susi	Analyst
Susi	Programmer

ABC Ename	ABC Eresp
Budi	DB Administrator
Busi	DB Administrator
Busi	Project Manager
Susi	DB Administrator

ABC Ename	ABC Etitle	ABC Eresp
Budi	Accountan	DB Administrator
Busi	Analyst	DB Administrator
Busi	Manager	DB Administrator
Busi	Analyst	Project Manager
Busi	Manager	Project Manager
Susi	Analyst	DB Administrator
Susi	Programmer	DB Administrator

5NF

ABC Ename	ABC Etitle
Budi	Accountan
Busi	Analyst
Busi	Manager
Susi	Analyst
Susi	Programmer

ABC Ename	ABC Eresp
Budi	DB Administrator
Busi	DB Administrator
Busi	Project Manager
Susi	DB Administrator

ABC Etitle	ABC Eresp
Accountan	DB Administrator
Analyst	DB Administrator
Manager	Project Manager
Programmer	DB Administrator

ABC Ename	ABC Etitle	ABC Eresp
Budi	Accountan	DB Administrator
Busi	Analyst	DB Administrator
Busi	Manager	Project Manager
Susi	Analyst	DB Administrator
Susi	Programmer	DB Administrator



Barangsiapa yang berangkat menimba ilmu untuk mengamalkan ilmu,
niscaya ilmu yang sedikit pun akan bermanfaat baginya